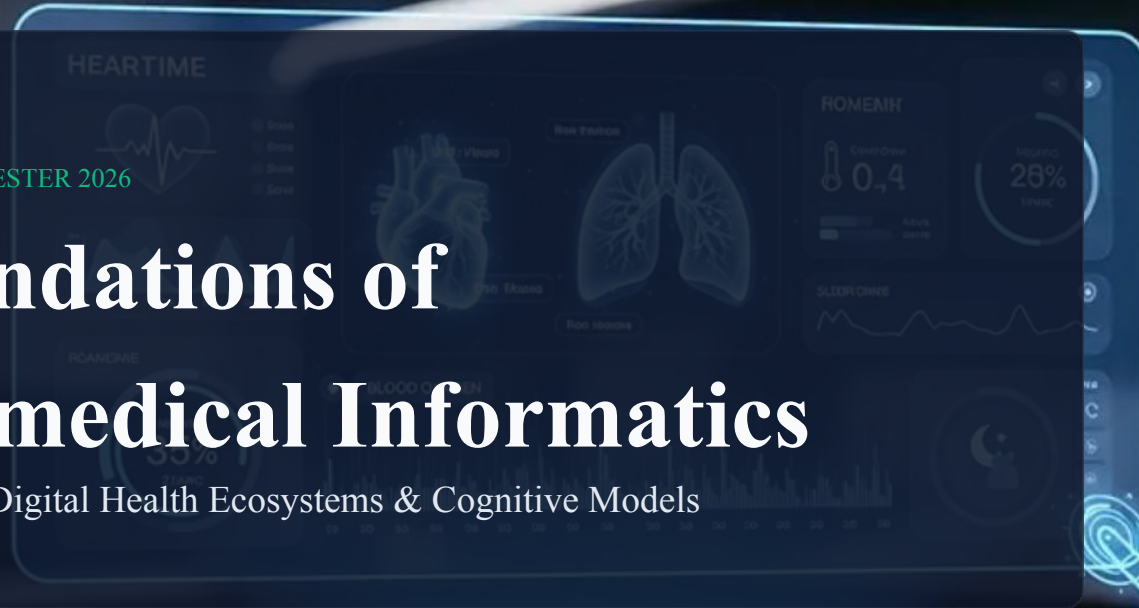


WINTER SEMESTER 2026

Foundations of Biomedical Informatics

Lecture 1: Digital Health Ecosystems & Cognitive Models



Today's Agenda

Introduction	Welcome and Introductions
Learning Objectives	What you are expected to learn
Course Structure	Major Topics covered in this course
Introduction to Biomedical Informatics	What, Why, How, Who
Pillars of Informatics	Cognition, Structure, Questioning, Searching, and Decision Making

Course Learning Objectives

Explain

foundational concepts of biomedical informatics and the learning healthcare system.

Evaluate

the design and functioning of complex biomedical information systems.

Apply

Apply different biomedical decision-making paradigms to clinical and public health use cases.

Design

Design AI-augmented clinical decision support systems with a focus on responsibility.

Goal

Understand biomedical informatics as an interdisciplinary field integrating **data, knowledge, and intelligent systems**

- Apply informatics approaches to **improve healthcare decision-making** in clinical and public health contexts
- Build foundations in **biomedical knowledge systems** (ontologies, terminologies, clinical knowledge representation)
- Learn key **health data standards and interoperability** frameworks for real-world integration
- Develop awareness of **privacy, security, and ethical responsibilities** in handling biomedical data
- Explore and evaluate **explainable AI (XAI)** methods for trustworthy decision support
- Gain skills to **design, evaluate, and responsibly deploy** informatics solutions in practice

Why this Course?

Biomedical Informatics: A Discipline Transcending Boundaries

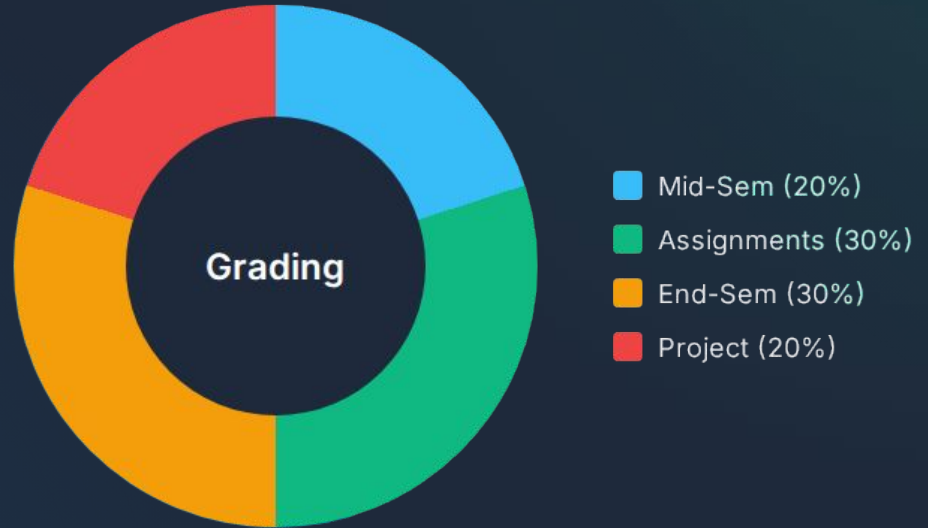
- **No Prerequisites:** No prior programming experience or high-level math skills required.
- **Thinking and Doing Focused.**
- **Hands-on Learning:** Step-by-step guidance through bioinformatics concepts without needing a biology degree.
- **Career Ready:** Your jumping-off point into computational and data science roles, which are in high demand across the healthcare industry.

Course Structure

Week Number	Lecture Topic	COs Met	Assignment/Labs/Tutorial
1	Learning Healthcare System	CO1	3+2 hours (Tutorial)
2	How Does Biomedical Informatics Differ from Computer Science Applied to Biomedicine?	CO1	3+2 hours (Exercises+Assignment-1)
3	Fundamentals of Data, Information, Knowledge	CO1, CO2	3+2 hours (Exercises)
4	Biomedical Information Systems and Medical Knowledge Management	CO1, CO2	3+2 hours (Quiz 1)
5	Privacy, Safety, Security of Biomedical Data	CO1, CO2, CO3, CO4	3+2 hours (Exercises)
6	Structured Data, Coding and Classification	CO1, CO2	3+2 hours (Exercises+Assignment-2)
7	Semistructured Data, Graphs, Unstructured Data	CO1, CO2	3+2 hours (Exercises)
8	Biomedical Decision Making - Cognitive Models	CO2, CO3	3+2 hours (Quiz 2)
9	Biomedical Decision Making - Knowledge Based Systems	CO2, CO3	3+2 hours (Exercises)
10	Biomedical Decision Making - Probabilistic Decision Support Systems	CO2, CO3	3+2 hours (Quiz 3)
11	Biomedical Decision Making - Machine Learning for Decision Support	CO2, CO3	3+2 hours (Exercises)
12	Biomedical Decision Making - AI Augmented Clinical Decision Support	CO2, CO3, CO4	3+2 hours (Exercises)
13	Explainability, Interpretability, Real World Usability of Applications	CO2, CO3, CO4	3+2 hours (Presentations)

Evaluation Scheme

Type of Evaluation	% Contribution in Grade
Mid-Sem	20
Assignments and Quizzes	30
End-Sem	30
Project	20



Project Guidelines

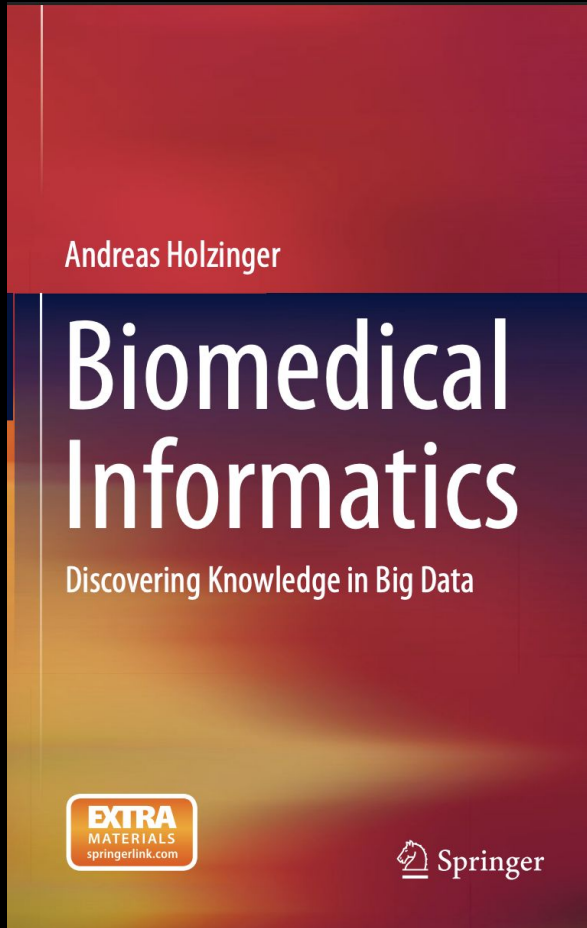
Learn to be effective in the real world by:

- Asking a strong, well-scoped question 
- Designing a **computational/programming/device-based approach** to answer it

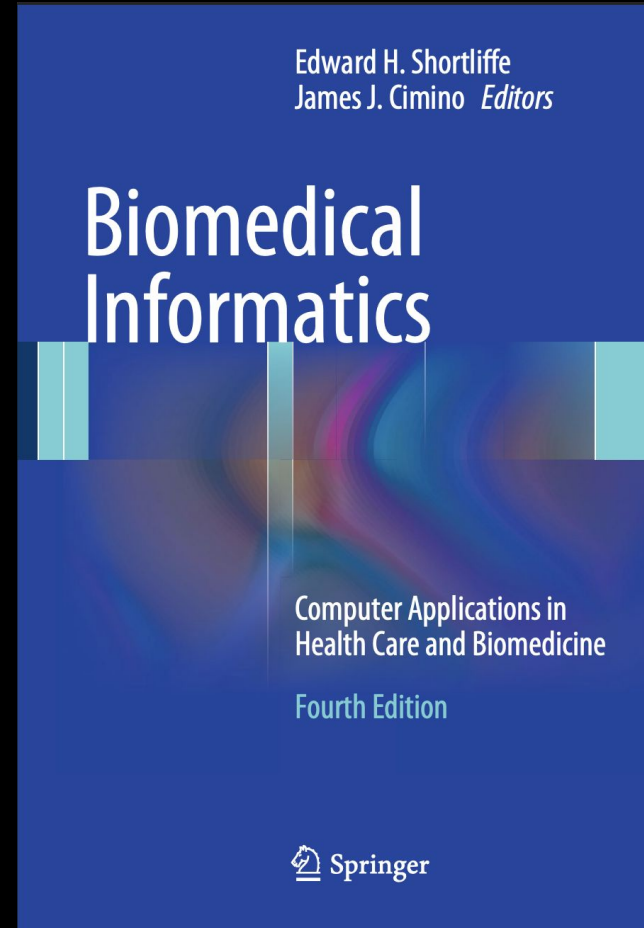
- Delivering a **final end-sem project + presentation** showcasing methods, results, and impact

Project modality: Team-based work in **groups of 4–6** (based on registrations)





Course Book



Reference Book



MODELS OF HEALTH

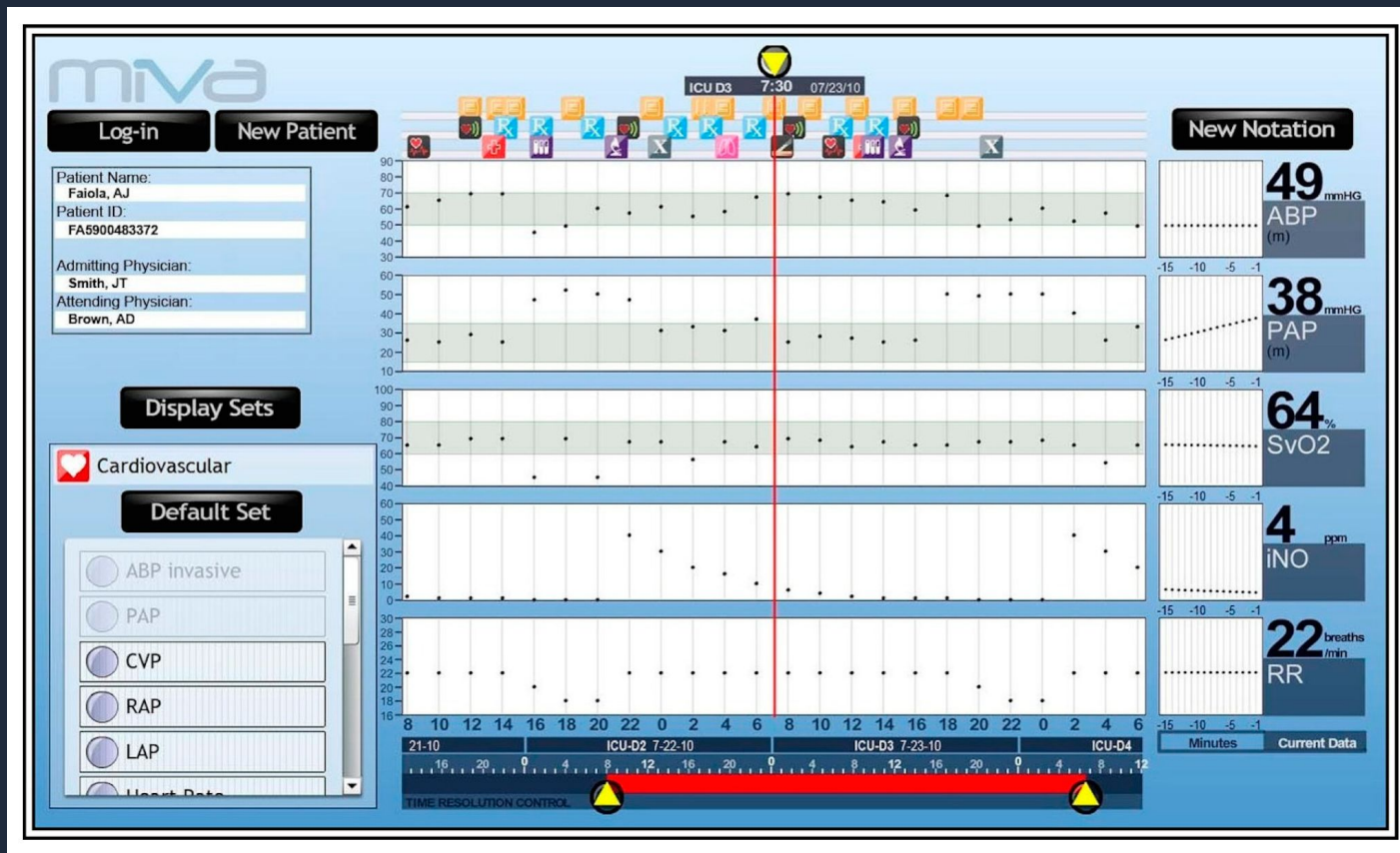
Bridging Data, Knowledge, and Intelligent Systems

Models of Information (Informatics)

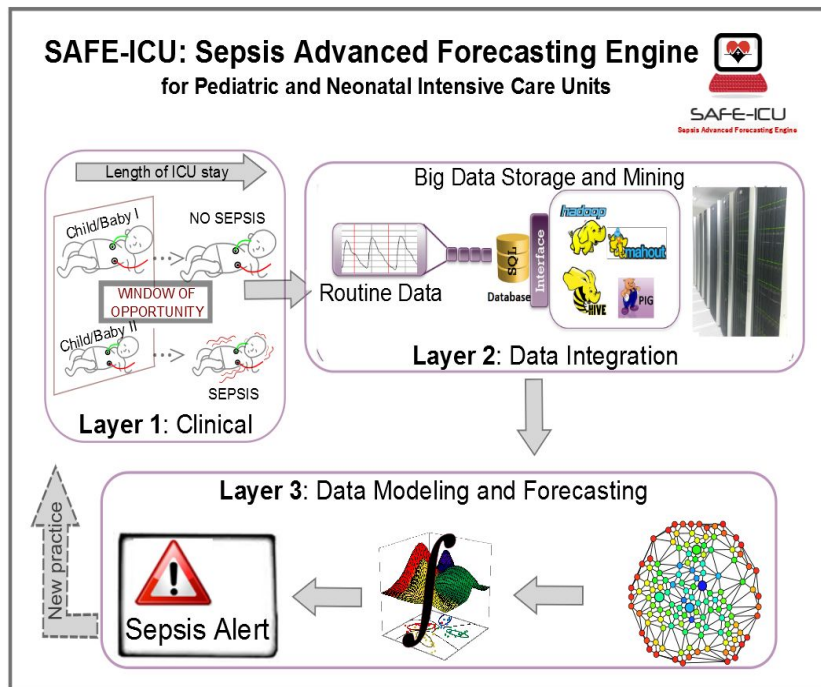
- **Informatics is "Map-Making":** We build simplified representations of complex clinical reality. However, *"A map is not the territory."*
- **Value > Perfection:** Models don't capture reality; they approximate it. Their value comes from being useful, not perfect.
- **The Stake:** In healthcare, poor models lead to wrong assumptions and wrong care. We must define "Data," "Information," and "Knowledge" precisely.

Discussion: What is a "model" you use every day—and where does it break?

Patient Timeline as a Mental Model



Example Designing Early Warning Systems for ICUs



Sethi T and Lodha R.

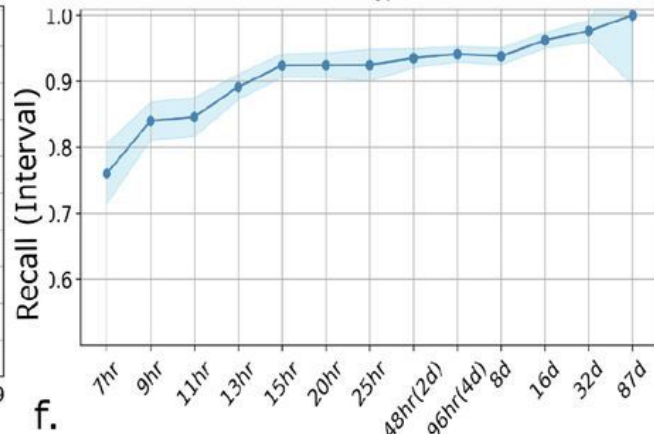
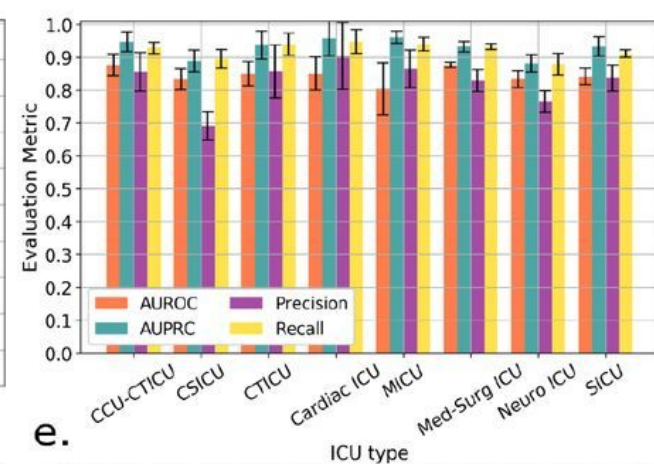
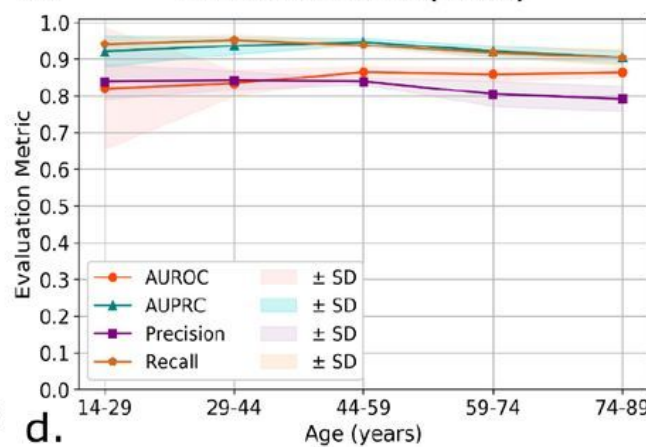
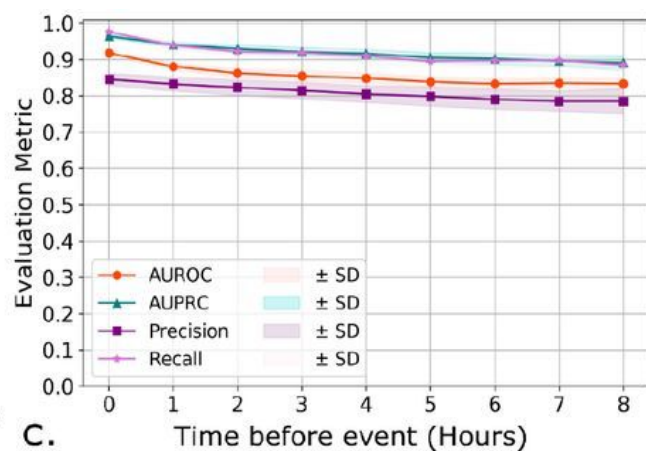
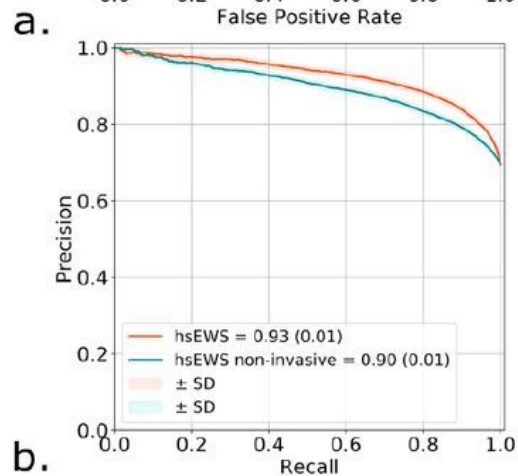
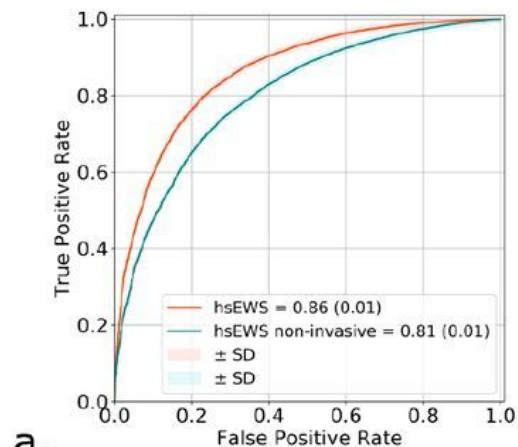
Early Detection of Sepsis in Paediatric Intensive Care Units through
Integration of Clinical and Big Data.

Wellcome Trust/DBT India Alliance Grant No. IA/CPHE/14/1/501504

Sepsis Advanced Forecasting Engine (SAFE-PICU)

>2.5M patient-hours of vitals data at
1-second resolution (and counting)

Integrate Machine Learning, Clinical
Knowledge to address gaps in critical
care



Nagori, Aditya et al. A Prospectively Validated Generalizable Model for Outcome Prognostication Using Shock Index in Intensive Care Units. Available at SSRN: <https://ssrn.com/abstract=4248013>

Why Should We Care?

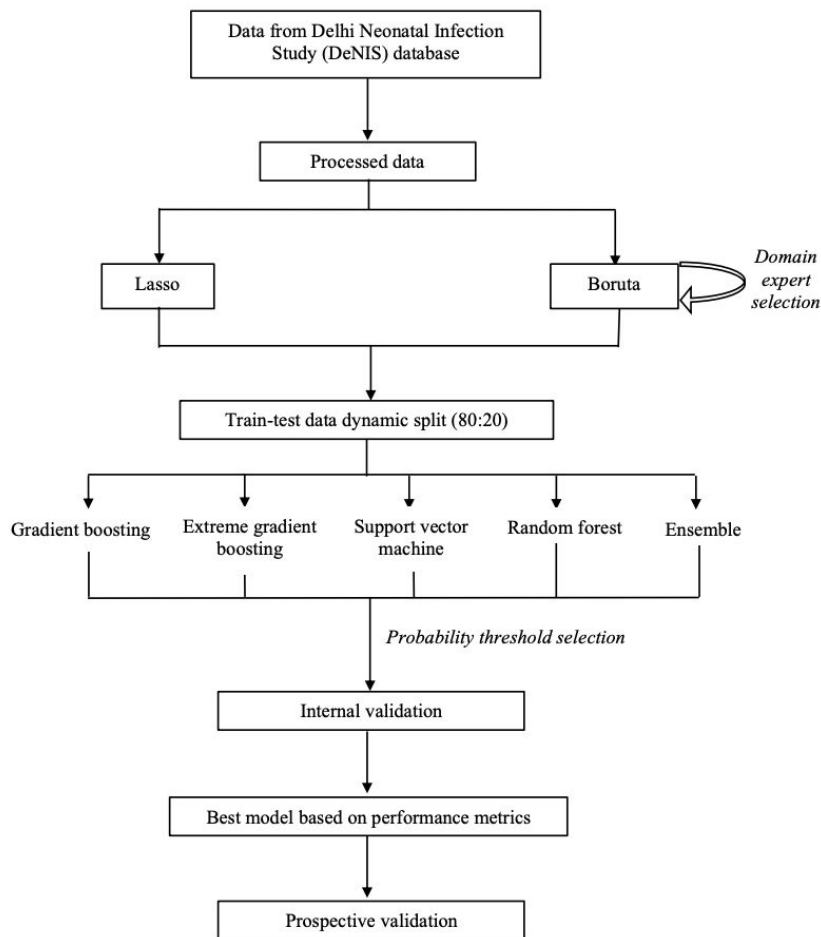
Outcome	Cohort	Odds ratio	95%CI	P-value
Mortality	eICU	15.99	12.92-19.69	<2.22e-16
	MIMIC-III	2.66	1.59-4.47	0.0001
	SAFE-ICU	2.61	0.82-7.9	0.09
LOS	eICU	4.72	4.48-4.97	< 2.220446e-16
	MIMIC-III	2.37	1.82-2.92	3.69648735e-16
	SAFE-ICU	7.79	4.89-10.69	2.7218e-07

Learning To Generalize to Low Resource Settings

5 min shock model (<i>Embeddings generated from 5-minute resolution data using a pre trained 5-minute language model and used as input to the shock model</i>) - Gradient Boost Model					
Metrics	Test Set: 5-min data using a 5-min LM	Test Set: 10-min data using a 5-min LM	Test Set: 15-min data using a 5-min LM	Test Set: 30-min data using a 5-min LM	Test Set: 60-min data using a 5-min LM
Accuracy	0.83 ± 0.004	0.84 ± 0.004	0.83 ± 0.004	0.79 ± 0.004	0.69 ± 0.005
Precision	0.88 ± 0.004	0.85 ± 0.004	0.83 ± 0.005	0.81 ± 0.005	0.85 ± 0.005
Recall	0.86 ± 0.005	0.92 ± 0.003	0.95 ± 0.003	0.90 ± 0.003	0.67 ± 0.006
F1-Score	0.87 ± 0.003	0.88 ± 0.003	0.88 ± 0.003	0.86 ± 0.003	0.75 ± 0.005
AUC	0.89 ± 0.004	0.88 ± 0.004	0.88 ± 0.004	0.83 ± 0.004	0.78 ± 0.005
AUPRC	0.94 ± 0.003	0.93 ± 0.003	0.93 ± 0.003	0.89 ± 0.004	0.87 ± 0.005

Early Warning Models Can Improve Rational Use

- **Data Source:** Delhi Neonatal Infection Study (2011–2014) across 3 level-3 neonatal units in India.
- Included: **Inborn neonates** with suspected EOS; excluded culture-negative cases.
- **Goal:** $\geq 90\%$ sensitivity to minimize missed sepsis cases.
- **Internal validation:** Sensitivity 90.3%, Specificity 40.6%.
- **External validation (n=147; 26 sepsis cases):**
 - Sensitivity: **92.3%**
 - Specificity: **37.2%**
 - PPV: **24.0%**, NPV: **95.7%**
 - **Asymptomatic neonates:** 100% sensitivity for perinatal-risk-only cases.
- **Projected Antibiotic reduction (What If Scenario):** 74.8% $\rightarrow$ 55.7% (–19.1%; 95% CI: –8.3 to –29.7).





शरीरमाद्यं खलु धर्मसाधनम्

Antimicrobial Stewardship Platform

आपका पूर्व-सूचक AI सहायक

Username

Password

☐ Remember Me

Not a member? [Register](#)

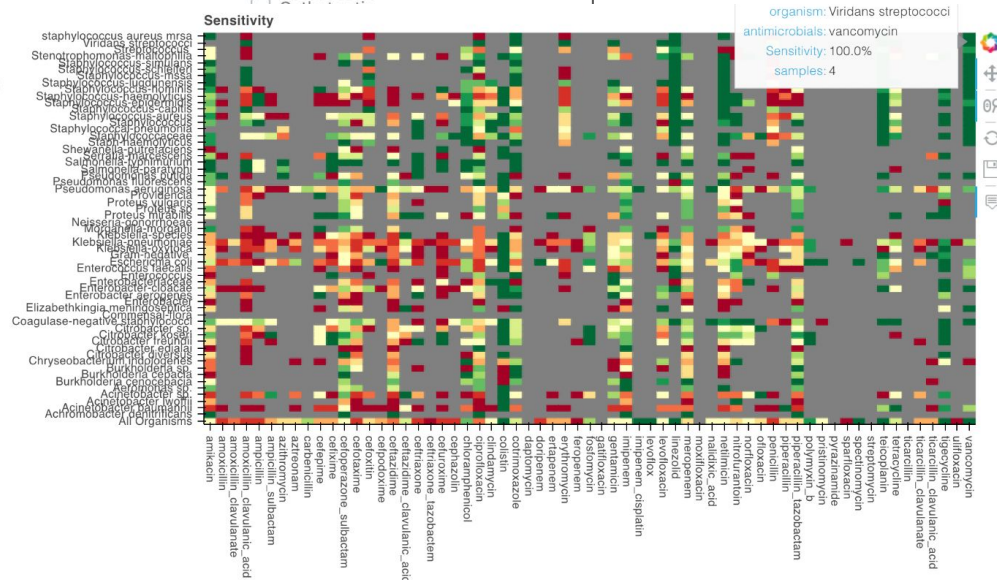
Login

- ☒ amikacin
- ☒ amoxicillin
- ☒ amoxicillin_clavulanate
- ☒ amoxicillin_clavulanic_acid
- ☒ ampicillin
- ☐ ampicillin_sulbactam
- ☐ azithromycin
- ☐ aztreonam
- ☐ carbenicillin
- ☐ cefepime
- ☐ cefixime
- ☐ cefoperazone_sulbactam
- ☐ cefotaxime
- ☐ ceftioxin
- ☐ cefpodoxime
- ☐ ceftazidime
- ☐ ceftazidime_clavulanic_acid
- ☐ ceftriaxone
- ☐ ceftriaxone_tazobactam
- ☐ cefuroxime
- ☐ cephalozin
- ☐ chloramphenicol

- ☒ *Achromobacter denitrificans*
- ☐ *Acinetobacter aeruginosa*
- ☒ *Acinetobacter baumannii*
- ☒ *Acinetobacter lwofii*
- ☐ *Acinetobacter* sp.
- ☒ *Aeromonas* sp.
- ☐ *Alcaligenes faecalis*
- ☒ Budding yeast
- ☐ *Burkholderia cenocepacia*
- ☐ *Burkholderia cepacia*
- ☐ *Burkholderia* sp.
- ☐ *Chryseobacterium indologenes*
- ☐ *Citrobacter diversus*
- ☐ *Citrobacter edialai*
- ☐ *Citrobacter freundii*
- ☐ *Citrobacter koseri*
- ☐ *Citrobacter* sp.
- ☐ Coagulase-negative staphylococci
- ☐ Commensal-flora
- ☐ *Elizabethkingia meningoseptica*
- ☐ *Enterobacter*

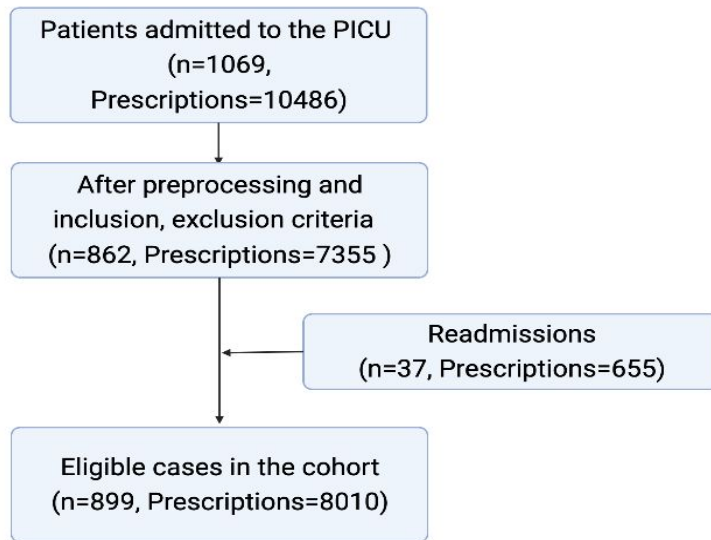
Sensitivity (%)

- ☐ Ascitic fluid
- ☐ Aspirate
- ☐ Bile
- ☐ Biopsy
- ☒ Blood
- ☐ Bone
- ☐ Bone marrow
- ☐ Broncho-Alveolar Lavage



Preventing Adverse Drug Drug Interactions in ICUs (and Hospitals)

- The retrospective data concurs with the background rate of PICU admission for both genders



Inclusion

Age **0 -17 years**

Stay **more than 24 hrs**

Medication **at least 2 drugs**

Exclusion

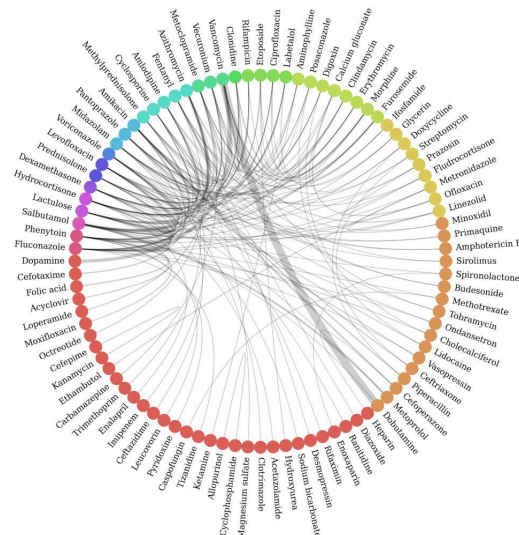
Topical medications were not considered

Most common drug combinations that are associated with pDDI in the SAFE-ICU

Nearly 50% (n=446) of those who were admitted to the ICU encountered at least one major, moderate, or minor interaction

Table 1: Frequencies of interacting drug combinations in PICU

Severity	Interacting drug pairs	Frequency	N(Patients)
Major	Fentanyl+Fluconazole	175	49
	Fluconazole+Midazolam	139	42
	Dexamethasone+Fentanyl	131	54
	Piperacillin+Vancomycin	107	34
	Fentanyl+Ondansetron	49	3
Moderate	Dobutamine+Vancomycin	469	151
	Dopamine+Vancomycin	170	87
	Azithromycin+Salbutamol	164	6
	Metoprolol+Salbutamol	85	1
	Amikacin+Vancomycin	78	35



Dashboard Overview

Upon launching the application, you will be presented with a user-friendly dashboard where you can:

CONTRAINDICATOR

Add TagConfirmReset ListSubmit

Upload a Treatment-Chart

Choose a file:

No file chosen

Upload

<https://github.com/tavlab-iiitd/ContraIndicator>

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Contraindicator

Settings

Search

Edit

Name

wq783e98fd/7

Diagnosis

Consultants

1

Fentanyl

2

Alvimopan

3

Atropine

4

Acarbose

5

Amcinonide

6

Albiglutide

7

Add Row

Add Box

Print

DDI

Add Cell

Drug-Drug Interaction Scanner

?

×

DDI Scanner

Potential Interactions:

	Drug A	Drug B	Interaction
1	Amcinonide	Albiglutide	Minor
2	Atropine	Fentanyl	Moderate
3	Acarbose	Fentanyl	Unknown
4	Acarbose	Albiglutide	Minor
5	Fentanyl	Alvimopan	Major



Questions?